Diabetic Patient Data Analytics Project Report

1. Project Overview

This project analyzes diabetic patient data to uncover patterns, risk factors, and healthcare insights that can help hospitals improve patient care and reduce readmission rates. The analysis uses the Diabetes 130-US Hospitals dataset (1999–2008) from Kaggle. The project follows a complete end-to-end analytics pipeline including data preprocessing, exploratory data analysis using Python, SQL-based data analysis, and dashboard visualization using Power BI.

Tools & Technologies Used:
- Python (Pandas, NumPy)
- MySQL (Data querying and analysis)
- Power BI (Interactive dashboard creation)
- Gamma (Presentation creation)

2. Dataset Summary

The dataset contains medical records of diabetic patients from 130 hospitals in the United States collected between 1999 and 2008. It includes demographic information, admission details, treatment intensity, medical tests, and readmission outcomes.

Key dataset characteristics:
- Patient demographics (age group, gender, race)
- Hospital admission type and discharge information
- Lab test results and treatment intensity
- Number of hospital encounters
- Readmission information (<30 days, >30 days, or no readmission)

3. Exploratory Data Analysis using Python

Exploratory Data Analysis (EDA) was conducted using Python libraries such as Pandas, NumPy to understand patterns in the dataset.

EDA tasks performed:
- Data loading using Pandas
- Handling missing values and inconsistent data
- Distribution analysis of patient demographics
- Correlation analysis between medical variables
- Visualization of age distribution, gender distribution, and treatment trends

Key Insights from EDA:
- Middle-aged and adult patients form the largest portion of diabetic admissions.
- Male patients are slightly higher in number compared to female patients.
- Certain treatment intensity levels correlate with higher readmission probability.

4. Data Analysis using SQL

After cleaning the dataset in Python, the processed data was loaded into MySQL for structured querying and analysis.

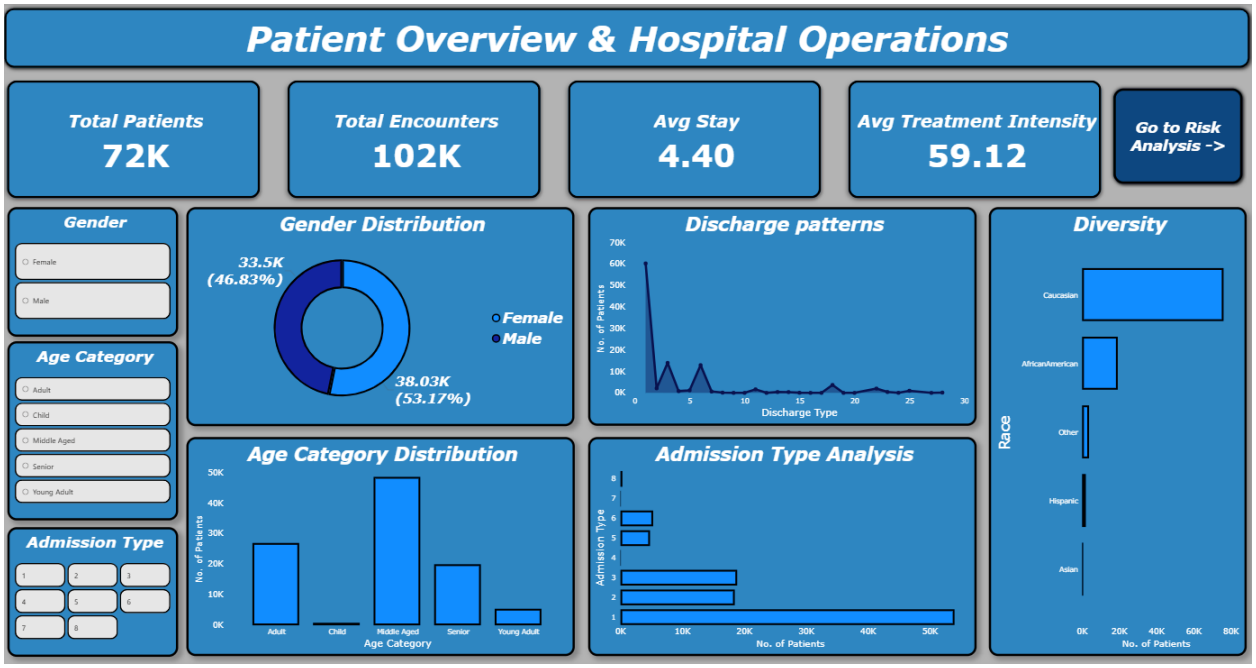
Example SQL analysis tasks:
- Calculate average hospital stay by age group
- Identify high-risk patients based on risk score
- Analyze readmission patterns
- Count patient distribution by admission type
- Evaluate treatment intensity levels across patient groups

SQL helped extract aggregated insights and supported the development of metrics used in the dashboard.

5. Dashboard in Power BI

A two-page interactive Power BI dashboard was created to visualize the analysis results and provide an intuitive interface for exploring hospital operations and patient risk insights.

Page 1: Patient Overview & Hospital Operations Dashboard



This page provides a high-level overview of hospital operations including total patients, encounters, average stay duration, and treatment intensity. It also visualizes patient distribution by gender, age category, admission type, discharge patterns, and race diversity.

Page 2: Readmission & Risk Analysis Dashboard

This page focuses on identifying high-risk patients and analyzing readmission patterns. It includes treatment effectiveness trends, readmission breakdown, clinical risk score distributions, and age groups most likely to experience readmission.

6. Recommendations

Based on the analysis, several recommendations can help healthcare providers improve diabetic patient management:

- Focus preventive care programs on middle-aged and adult patients who represent the largest risk group.
- Monitor patients with high treatment intensity closely to reduce the likelihood of readmission.
- Implement targeted care plans for high-risk patients identified through risk scoring models.
- Improve follow-up care within 30 days after discharge to reduce readmission rates.
- Use predictive analytics to identify potential complications before they lead to hospital readmission.